

EEG Signal Collection & Random Forest-Based Imputation

Complete Question & Answer Guide

Everything Explained — From Zero Knowledge to Full Understanding

Aditya Krishna | Roll No: 225CS2009

Guide: Prof. Pabitra Mohan Khillar

Department of Computer Science & Engineering

National Institute of Technology, Rourkela

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SECTION 1: Introduction to EEG — The Basics

Q1. What is EEG? Explain in simple terms.

EEG stands for **Electroencephalography**. Let us break the word down: Electro = electricity, Encephalo = brain, Graphy = recording. So EEG simply means **recording the electrical activity of the brain**.

Your brain is made up of billions of nerve cells called **neurons**. These neurons constantly communicate with each other by sending tiny electrical signals. When millions of neurons fire together, they produce a small electrical voltage that can be detected on the surface of your scalp (the skin on top of your head).

EEG captures these tiny voltage changes using small metal sensors called **electrodes** placed on the scalp. The signals are measured in **microvolts (μV)** — one microvolt is 0.000001 volts, which is about one million times smaller than a standard AA battery.

Note: EEG does NOT use any radiation, injections, or invasive procedures. It is completely safe and painless.

Q2. Why is EEG important? What are its main uses?

EEG is one of the very few tools that can observe what the brain is doing **in real time**, with millisecond precision. This makes it extremely valuable for both medical and research purposes.

Medical Uses:

- **Epilepsy Diagnosis:** EEG is the gold standard test for epilepsy. It detects the abnormal electrical discharges (spikes) that cause seizures.
- **Sleep Studies:** Doctors use EEG to identify sleep disorders like insomnia, sleep apnea, and narcolepsy by analyzing brain wave patterns during sleep.
- **Brain Injury Assessment:** After a stroke or head injury, EEG shows which parts of the brain are affected.
- **Coma Monitoring:** EEG tells doctors how deep a coma is and whether the brain is still active.

Research and Technology Uses:

- **Brain-Computer Interface (BCI):** Allows paralyzed patients to control computers or robotic arms just by thinking.
- **Mental Health Research:** Studying ADHD, depression, and anxiety through their unique brain patterns.
- **Cognitive Research:** Understanding how the brain processes learning, memory, attention, and emotions.

Q3. What is a microvolt (μV)? Why are EEG signals so small?

A **microvolt (μV)** is a unit of electrical voltage. The prefix 'micro' means one-millionth. So 1 μV = 0.000001 volts.

EEG signals are tiny because you are not recording directly from neurons. You are recording through the scalp, skull bone, and brain fluid — all of which act like barriers that weaken the signal significantly. By the time the brain's electrical activity reaches the scalp, it has been reduced to just 10–100 μV.

To put this in perspective: a standard household electrical outlet is 220,000,000 μV (220 volts). The EEG signal is roughly **two million times weaker** than household electricity. This is why EEG needs a powerful amplifier to make the signal measurable.

Note: This tiny signal also means EEG is very sensitive to noise. Muscle movements, eye blinks, and even electrical devices nearby can interfere with the recording.

Q4. What is temporal resolution and why does EEG have good temporal resolution?

Temporal resolution means how precisely a tool can measure changes over time. A tool with high temporal resolution can detect events that happen very quickly.

EEG has **millisecond-level temporal resolution**, meaning it can detect brain events that last as short as 1/1000th of a second. This is important because neurons fire extremely fast — a single thought or reaction can involve brain activity lasting just a few milliseconds.

By comparison, **fMRI (Functional MRI)** — another popular brain imaging tool — has very poor temporal resolution (events take 1–2 seconds to show up), though it has much better spatial resolution (it can precisely locate WHERE in the brain activity is happening). EEG is better for WHEN; fMRI is better for WHERE.

SECTION 2: Brain Frequency Waves

Q5. What are brain waves? How are they classified?

Brain waves are patterns of electrical activity in the brain. When neurons fire repeatedly in a synchronized rhythm, they create waves — just like ripples in water. We classify these waves by their **frequency** (how many complete cycles happen per second), measured in **Hertz (Hz)**.

1 Hz means 1 complete cycle per second. 30 Hz means 30 complete cycles per second. Higher frequency = faster oscillation = more active brain state.

There are 5 main frequency bands in EEG, each linked to a different mental state:

Q6. What is the Delta wave (0.5–4 Hz)? When does it occur?

Delta waves are the **slowest brain waves**, oscillating between 0.5 and 4 times per second. They have the highest amplitude (tallest waves) of all brain wave types.

Delta waves are most prominent during **deep, dreamless sleep** — the kind of sleep where you are completely unconscious and very difficult to wake up. This is the most restorative stage of sleep where the body repairs itself.

In adults who are awake, high Delta activity is abnormal and can indicate **brain injury, tumors, or severe neurological disorders**. In infants, Delta is normal even during wakefulness since their brains are still developing.

Note: In this project's dataset, Delta had the highest mean power (60.45 μV^2) but also the highest standard deviation (273.78), indicating large bursts of activity across different subjects and tasks.

Q7. What is the Theta wave (4–8 Hz)? When does it occur?

Theta waves oscillate 4 to 8 times per second. They are associated with **drowsiness, light sleep, meditation, deep relaxation, and creative states**.

Theta is also very active when you are in a **hypnagogic state** (the drowsy state just before falling asleep) and during **REM sleep** (the stage where you dream). Interestingly, Theta also appears during focused internal tasks like daydreaming and accessing deeply stored memories.

In clinical settings, excess Theta activity while awake is associated with **ADHD** (children with ADHD show high Theta, low Beta), and with **Alzheimer's disease** where brain networks slow down.

Q8. What is the Alpha wave (8–13 Hz)? When does it occur?

Alpha waves oscillate 8 to 13 times per second. They are the brain's '**idle**' **rhythm** — present when you are awake but calm, relaxed, and not actively thinking about anything specific.

Close your eyes and relax right now. Your Alpha waves will immediately increase, especially over the back of your head (occipital region). Open your eyes and they will drop. This is called the **Alpha Blocking response** and is one of the most classic findings in EEG.

Alpha waves are important for **anxiety management** — people with anxiety tend to have lower Alpha activity, which is why meditation and mindfulness practices (which boost Alpha) help reduce anxiety.

- **Alpha Asymmetry:** Depression is linked to lower Alpha in the left frontal region compared to the right. This asymmetry is a potential biomarker for depression.

Q9. What is the Beta wave (13–30 Hz)? When does it occur?

Beta waves oscillate 13 to 30 times per second. They are the brain's **active thinking mode** — dominant when you are alert, focused, problem-solving, making decisions, or engaged in conversation.

Beta is the state you are in right now while reading and concentrating. Beta activity increases when you are nervous or anxious (because the brain is highly alert) and decreases during relaxation and meditation.

- **High Beta** (over 25 Hz) is associated with anxiety, stress, and hypervigilance.
- **Low Beta** (13–15 Hz, called SMR or Sensorimotor Rhythm) is associated with calm focused attention and is targeted in neurofeedback training for ADHD.

Q10. What is the Gamma wave (30–45 Hz)? When does it occur?

Gamma waves are the **fastest brain waves** we commonly measure in EEG, oscillating 30 to 45 times per second (some researchers extend this up to 100 Hz).

Gamma is linked to **higher cognitive functions**: complex thought, binding information from different brain regions, consciousness, and perception. When you have a sudden 'aha' moment or insight, there is typically a burst of Gamma activity.

Long-term meditators (like Buddhist monks) show extremely high Gamma activity even at rest. Research suggests this is associated with heightened awareness and compassion.

Note: Gamma is also the most difficult band to measure reliably because high-frequency muscle activity (from jaw clenching, facial movements) overlaps with Gamma frequencies and can contaminate the signal.

Q11. What is frequency in Hz and what does Welch's PSD method mean?

Frequency means how often something repeats per second. If a wave completes 10 full cycles in one second, its frequency is 10 Hz.

PSD stands for Power Spectral Density. It tells you how much power (energy) exists at each frequency in a signal. Think of it like a music equalizer — it shows you how much bass (low frequencies), midrange, and treble (high frequencies) are present.

Welch's Method is a popular way to calculate PSD reliably. Instead of computing PSD on the entire signal at once (which gives noisy results), Welch's method:

- Divides the signal into overlapping windows (in this project: 4-second windows with 50% overlap)
- Applies a **Hann window** (a mathematical shape that smooths the edges of each segment to reduce distortion)
- Computes the PSD for each window separately
- Averages all the PSDs together to get a smooth, reliable estimate

The result is a curve showing power (in $\mu\text{V}^2/\text{Hz}$) at each frequency from 0 to 128 Hz. You then sum up the power in specific frequency ranges to get your five band powers (Delta, Theta, Alpha, Beta, Gamma).

SECTION 3: Electrodes and the 10-20 System

Q12. What are EEG electrodes? What are gold-cup electrodes?

EEG **electrodes** are small metal sensors that detect the tiny electrical voltages on the scalp. They act as the interface between the biological brain and the electronic recording equipment.

Gold-cup electrodes are cup-shaped sensors made of pure gold (or gold-plated metal). The cup shape holds **conductive gel** — a special paste that improves electrical contact between the electrode and the scalp skin.

Why gold?

- Gold does not rust or corrode, ensuring stable, consistent measurements over time.
- Gold has excellent electrical conductivity.
- Gold is biocompatible — it does not irritate the skin.

Without the conductive gel, the dry skin acts as a high resistance barrier and the signal quality is very poor. The gel fills in microscopic air gaps between the electrode and skin.

Q13. What is the International 10-20 System? Why is it called 10-20?

The **International 10-20 System** is a standardized method for placing EEG electrodes on the scalp so that every lab in the world uses the same positions. This ensures that EEG data from different hospitals and countries can be compared.

It is called **10-20** because electrode positions are spaced either **10%** or **20%** of the total front-to-back or left-to-right distance of the skull. This proportional spacing means the system works for heads of any size — whether you have a small or large head, the electrodes land in the same brain regions.

Electrode Naming Convention:

- **Fp** = Frontal Pole (forehead area)
- **F** = Frontal lobe (planning, decision making)
- **C** = Central (motor/sensory strip)
- **P** = Parietal (spatial awareness, sensory integration)
- **O** = Occipital (visual processing)
- **T** = Temporal (hearing, memory, language)
- **Odd numbers** (1, 3, 5, 7) = Left hemisphere
- **Even numbers** (2, 4, 6, 8) = Right hemisphere
- **z suffix** (Fz, Cz, Pz) = Midline (centre of the head)

Q14. In this project, which electrodes were used and why?

This project used 5 electrode positions: **Fp1, Fp2, Cz, O1, and O2**, plus a reference on the left earlobe and ground on the right earlobe.

- **Fp1 and Fp2** (Frontal Pole left and right): Capture frontal lobe activity associated with cognitive tasks like learning and mathematics.
- **Cz** (Central midline): Captures central activity related to sensorimotor processing.
- **O1 and O2** (Occipital left and right): Capture visual cortex activity and strong Alpha rhythm.

Why Reference and Ground Electrodes?

EEG measures voltage **difference** between two points. The **reference electrode** (left earlobe, A1) provides a 'neutral' comparison point. The **ground electrode** (right earlobe, A2) helps eliminate common-mode electrical noise (like 50 Hz mains interference).

Note: The ear lobes are chosen as reference/ground because they are far from the brain and have relatively low brain electrical activity.

Q15. What are the three electrodes in the basic setup (Active, Reference, Ground)?

In a basic 3-electrode EEG setup, each electrode plays a specific role:

- **Active Electrode:** Placed on the brain region you want to study (e.g., Fp1 on the forehead). This picks up the brain signal plus all electrical noise in the environment.
- **Reference Electrode:** Placed on an electrically quiet location (e.g., earlobe). This also picks up environmental noise, but almost no brain signal.
- **Ground Electrode:** Connected to the amplifier's ground point. It provides a stable electrical baseline, preventing the amplifier from being overwhelmed by common noise.

The amplifier computes **Active minus Reference**. Since both electrodes pick up the same environmental noise, subtracting them cancels the noise and leaves only the brain signal. This is called **differential amplification**.

SECTION 4: Hardware, Data Collection, and Signal Processing

Q16. What is an Arduino Mega 2560? Why was it used for EEG?

An **Arduino Mega 2560** is a small, affordable microcontroller board — essentially a tiny computer on a circuit board. It has a processor, memory, and a set of input/output pins that can read sensors and communicate with computers.

In this project, the specific board used is the **EWS All-in-one Arduino Mega IoT Grove Kit**, which is the Arduino Mega with several built-in communication modules:

- **Wi-Fi module** for wireless internet connection
- **Bluetooth module** for short-range wireless
- **SIM7672S Cellular modem** for mobile network (GSM/LTE) connectivity
- **ZigBee module** for low-power mesh networking

The Arduino was chosen because it is **affordable (under Rs. 3,000)**, widely available in India, supports the Grove ecosystem of sensors, and the same board can later be upgraded to wireless transmission without hardware changes.

Q17. What is an ADC? What does 10-bit ADC mean?

An **ADC (Analog-to-Digital Converter)** converts continuous analog signals (like the varying voltage from an EEG electrode) into discrete digital numbers that a computer can process.

Think of it like this: the real EEG signal is a smooth, continuously varying voltage. But a computer can only work with numbers (0s and 1s). The ADC's job is to sample the voltage at regular intervals and convert each measurement to a number.

10-bit refers to the resolution: the ADC represents each sample as a 10-digit binary number. This gives $2^{10} = 1,024$ **discrete levels**. For example, if the signal range is 0–5 volts, each step = $5V/1024 \approx 4.88$ mV per step.

Note: Higher bit depth = finer resolution. Professional EEG systems use 24-bit ADCs (16 million levels), which is why they capture more detail. The 10-bit ADC in this project is lower resolution but sufficient for band-power analysis.

Q18. What is sampling rate? Why was 256 Hz chosen?

Sampling rate (also called sampling frequency) is how many times per second the ADC measures the signal. A rate of 256 Hz means the signal is measured 256 times every second.

The key rule governing sampling rate is the **Nyquist Theorem**: to accurately capture a signal, you must sample at **at least twice** the highest frequency you care about.

Since EEG signals of interest go up to about 45 Hz (Gamma band), we need at least $2 \times 45 = 90$ Hz sampling rate. Using 256 Hz gives comfortable margin — it captures everything up to 128 Hz accurately, well above the 45 Hz Gamma band.

- 256 Hz is also a **power of 2** (2^8), which makes mathematical operations like FFT (Fast Fourier Transform) computationally efficient.
- It is the most commonly used sampling rate in research-grade EEG systems.

Q19. What is an Instrumentation Amplifier (INA128)? Why is it needed?

An **instrumentation amplifier** is a special type of amplifier designed to amplify very small differential signals while rejecting common-mode noise. The **INA128** is a specific chip made by Texas Instruments.

The EEG signal arriving at the scalp is only 10–100 μV . Before the ADC can process it, it needs to be amplified to a usable voltage range (0–5V for the Arduino). A gain of approximately **1000×** means a 100 μV signal becomes 100 mV = 0.1V, which the ADC can measure.

Key Properties of INA128:

- **High Common-Mode Rejection Ratio (CMRR):** Rejects 50 Hz mains hum and other noise common to both electrodes.
- **High input impedance:** Does not load the electrode, preserving signal integrity.
- **Low noise:** Adds minimal noise of its own.
- **Adjustable gain:** One resistor controls amplification (gain $\approx$ 1000 in this project).

Q20. What does band-pass filtering (0.5–45 Hz) do?

A **band-pass filter** is an electronic or mathematical filter that allows only a specific range of frequencies to pass through and blocks everything outside that range.

Setting the range to **0.5–45 Hz** means:

- **High-pass at 0.5 Hz:** Blocks DC offset and very slow drifts caused by electrode movement, sweat, or skin potential changes. Without this, the signal would slowly wander up and down, making analysis impossible.
- **Low-pass at 45 Hz:** Blocks high-frequency noise above 45 Hz, including muscle artifacts (EMG noise from facial muscles, jaw clenching) and electrical interference. Since EEG brain signals do not exceed 45 Hz in most analyses, this filtering removes noise without losing useful information.

In this project, a **4th-order zero-phase Butterworth filter** was used. 'Zero-phase' means the filter does not shift the signal in time (no phase distortion). '4th-order' means the filter has a steeper rolloff — it more aggressively cuts frequencies outside the band.

Q21. What is a 50 Hz notch filter and why is it needed?

In India (and most of the world), electrical power lines operate at **50 Hz**. All electrical devices connected to the grid (lights, computers, chargers) constantly emit 50 Hz electromagnetic interference that is picked up by EEG electrodes.

This 50 Hz noise is called **mains hum** or **power line interference**. It is often the strongest signal in the recording, completely masking the brain signals.

A **notch filter** (also called a band-stop filter) eliminates a very narrow frequency range. A 50 Hz notch filter removes **only** the 50 Hz component while leaving all other frequencies untouched. The quality factor $Q=30$ means the notch is very narrow (± 1.67 Hz around 50 Hz), so it does not distort neighboring frequencies.

Note: In the USA, power lines run at 60 Hz, so American EEG systems use a 60 Hz notch filter instead.

Q22. What is artefact rejection? What are motion artefacts?

An **artefact** in EEG is any signal that does not come from the brain — it is contamination from non-neural sources. Artefact **rejection** means detecting and removing these contaminated portions from the data.

Common EEG Artefacts:

- **Motion artefacts:** When the subject moves, electrodes shift on the scalp, causing large voltage spikes. These look very different from brain waves — they are much larger ($>150\ \mu\text{V}$) and have sharp, sudden onsets.
- **Eye blink artefacts:** Eye blinks and eye movements create large electrical signals (electrooculogram, EOG) that spread to frontal electrodes.
- **Muscle artefacts (EMG):** Jaw clenching, swallowing, or facial muscle activity produces high-frequency contamination.
- **ECG artefacts:** The heartbeat creates regular electrical pulses that can appear in EEG.

In this project, any sample exceeding $\pm 150\ \mu\text{V}$ is considered a motion artefact. Since brain signals rarely exceed $100\ \mu\text{V}$ under normal conditions, anything above $150\ \mu\text{V}$ is almost certainly artefact. These samples are removed and replaced by **linear interpolation** — a straight line is drawn between the clean samples on either side of the artefact.

SECTION 5: Datasets

Q23. What is the NIT Rourkela dataset collected in this project?

This is an **original dataset** recorded specifically for this project at NIT Rourkela using the low-cost Arduino-based system. It contains EEG band-power features from real subjects performing different cognitive tasks.

Dataset Summary:

- **30 subjects** — students and participants from NIT Rourkela
- **117 CSV source files** (multiple sessions per subject)
- **7,787 total records** — each record represents one second of EEG
- **4 brain states recorded:** Learning, Maths problem-solving, Meditation, Song Listening
- **5 features per record:** Delta, Theta, Alpha, Beta, Gamma band powers
- **Sampling:** 1 Hz epoch rate (one row of features per second)

The high standard deviation values (especially Delta: 273.78 and Theta: 356.95) indicate that these bands show large transient bursts across different tasks — confirming that the dataset captures genuine brain state variation.

Q24. What is the Kaggle BRMH dataset?

The **BRMH (Brain Resting-state Mapping of Health) dataset** is a publicly available clinical EEG dataset on Kaggle. It contains EEG recordings from patients diagnosed with various psychiatric and neurological disorders.

Key Statistics:

- **945 patients** with clinical EEG recordings
- **19-channel EEG** using the full 10-20 placement system
- **6 frequency bands:** Delta, Theta, Alpha, Beta, High-Beta, Gamma
- **Features:** Absolute band power + Coherence values = 1,000+ feature columns

Disorders Covered:

- Addictive disorders (Alcohol use disorder)
- Anxiety disorders, Depression, PTSD
- Trauma and stress-related disorders
- Schizophrenia and mood disorders
- Healthy control subjects

This dataset is valuable for future work in **disorder classification** — after imputing missing values, machine learning models can be trained to classify which disorder a patient has based on their EEG band pattern.

Q25. What is the Temple University Hospital EEG Corpus?

The **Temple University Hospital (TUH) EEG Corpus** is one of the largest publicly available clinical EEG datasets in the world. It was compiled at Temple University Hospital in Philadelphia, USA, and is widely used for epilepsy research.

Key Facts:

- **Over 10,000 EEG recordings** from real hospital patients
- Contains annotated **seizure events** with precise start and end times
- Includes multiple seizure types: focal, generalized, absence seizures
- Long recordings (hours) with many instances of electrode dropout and missing data

This dataset is particularly relevant to this project because it demonstrates the real clinical need for missing data imputation. Long hospital recordings frequently have corrupted segments due to electrode displacement during sleep, movement, or patient care activities.

The HTI-MissForest pipeline developed in this project could directly improve the quality of TUH dataset preprocessing before seizure detection algorithms are applied.

SECTION 6: Missing Data and Imputation Techniques

Q26. What is missing data in EEG? What causes it?

In a perfect world, every EEG recording session would produce a complete, clean dataset. In reality, **missing data** (gaps in the recording where no valid signal exists) is very common. These gaps appear as blank or corrupted entries in the dataset.

Causes of Missing EEG Data:

- **Electrode lifting:** The subject moves, causing a gold-cup electrode to lose contact with the scalp. The signal drops completely until the electrode is reattached.
- **Motion artefacts:** After artefact rejection, the corrupted segments are replaced with NaN (Not a Number) values — creating intentional gaps.
- **Serial communication loss:** In low-cost systems like this project, occasionally a data packet does not transmit correctly over USB, leaving a missing row.
- **Conductive gel drying:** Over long recordings (hours), the gel can dry out, increasing electrode impedance and degrading signal quality.
- **Sensor noise bursts:** Random electronic noise can temporarily overwhelm the signal.

Q27. What is MCAR? Why is the evaluation protocol important?

MCAR stands for Missing Completely At Random. This is one of three categories of missing data:

- **MCAR:** The missingness has no relationship to any data values. A value is missing purely by chance. Example: randomly dropping 5% of records.
- **MAR (Missing At Random):** The missingness depends on other observed variables but not on the missing value itself.
- **MNAR (Missing Not At Random):** The missingness depends on the value that is missing. Example: high-Delta periods are more likely to be corrupted.

This project uses **MCAR for evaluation**: 5% of real values are randomly removed, imputation is run, and errors are measured only at those removed positions. This is the most **honest benchmark** because:

- The true values are known (they were removed artificially), so error can be computed precisely.
- MCAR does not give artificial advantages to any imputation method.
- The evaluation is reproducible because a fixed random seed (42) controls which values are removed.

Q28. What is Mean Imputation? What are its problems?

Mean imputation is the simplest approach: replace every missing value with the **average (mean)** of that column across all non-missing rows.

For example, if the Alpha band has values [10, 15, NaN, 12, 14] and the mean of the non-missing values is 12.75, then the NaN is replaced with 12.75.

Problems with Mean Imputation:

- **Distorts variance:** Adding many identical values (the mean) artificially reduces the spread of the data. Statistics like standard deviation become artificially low.
- **Ignores correlations:** It does not use information from other bands. If Alpha is high, Beta is usually lower — mean imputation ignores this relationship entirely.
- **Distorts frequency structure:** In EEG, consecutive time points are correlated. Inserting the mean disrupts the time-series structure.

Q29. What is KNN Imputation? Why does it fail for EEG?

KNN (K-Nearest Neighbors) Imputation finds the K most similar rows (neighbors) in the dataset and uses the average of those neighbors' values to fill in the missing entry.

'Similar' is defined by Euclidean distance — rows that have close values across all observed features are considered neighbors.

Why it struggles with EEG band-power data:

- **Inter-band correlations:** EEG bands are correlated in complex, non-linear ways. Alpha and Beta are anticorrelated (one goes up as the other goes down). Simple distance metrics in KNN do not capture these relationships well.
- **High dimensionality:** When there are 5+ interrelated columns, finding meaningful neighbors in Euclidean space becomes unreliable.
- **No temporal awareness:** KNN does not use the time-series structure of the data.

Q30. What is MICE? What does it stand for?

MICE stands for Multiple Imputation by Chained Equations. It is a sophisticated statistical method that imputes missing values by building a regression model for each incomplete column.

How MICE works, step by step:

- **Step 1 — Initialize:** Fill all missing values with simple estimates (e.g., column means) as a starting point.
- **Step 2 — For each incomplete column:** Treat that column as the target variable (y). Use all other columns as predictor variables (X). Train a regression model on rows where y is observed. Use that model to predict the missing values in y.
- **Step 3 — Repeat:** Cycle through all incomplete columns, one by one. Each pass is called one 'iteration'. The estimates improve with each iteration because each model benefits from the improved estimates of other columns.
- **Step 4 — Convergence:** Stop when the estimates stop changing significantly between iterations (convergence).

The key advantage of MICE is that it uses **all available information** from all other variables to estimate each missing value, properly capturing correlations between variables.

SECTION 7: Random Forest and the HTI-MissForest Pipeline

Q31. What is a Decision Tree? (Foundation of Random Forest)

A **Decision Tree** is a machine learning model that makes predictions by asking a series of yes/no questions about the input data, following a tree-like branching structure.

Imagine you want to predict if a patient has epilepsy based on their EEG features. A decision tree might ask:

- Is Delta power $> 100 \mu V^2$? → If YES: Is Theta power $> 50 \mu V^2$? → If YES: Likely epilepsy.
- Is Alpha power $< 5 \mu V^2$? → If YES: Is Beta power $> 30 \mu V^2$? → Likely anxiety.

The tree learns these rules automatically from training data. Each branching point is called a **node**, and the final answers at the bottom are called **leaves**.

A single decision tree can overfit — it memorizes the training data too well and performs poorly on new data. This is why we use **Random Forest** instead.

Q32. What is a Random Forest? How does it work?

A **Random Forest** is an **ensemble** of many decision trees working together. Instead of trusting one tree's prediction, it builds many trees and combines their predictions by averaging (for regression) or majority voting (for classification).

How it creates diversity among trees:

- **Bootstrap sampling:** Each tree is trained on a random sample of the training data (with replacement). Some rows appear multiple times, others not at all.
- **Random feature selection:** At each branching point, each tree only considers a random subset of features to split on. This ensures trees are different from each other.

The key insight: individual trees make errors, but their errors are **random and uncorrelated**. When you average many different trees, the errors cancel out and the correct signal amplifies. This is called the **Wisdom of Crowds** effect.

In this project, the Random Forest uses **50 trees** (50 estimators) and each tree has a **maximum depth of 10** levels (to prevent overfitting).

Q33. What is the HTI-MissForest pipeline? Explain each step in detail.

HTI-MissForest is the novel imputation pipeline developed in this project. HTI stands for **Hybrid Temporal Imputation**. Each step is designed to improve imputation quality:

Step 1: Feature Engineering

Three new columns are added to the 5-band table before imputation:

- **Total Power** = Delta + Theta + Alpha + Beta + Gamma. Represents overall brain activity level.
- **Alpha/Beta Ratio** = Alpha ÷ Beta. This ratio captures attentional state — high ratio means relaxed, low ratio means focused. Alpha and Beta are known to be anticorrelated.
- **Theta/Gamma Ratio** = Theta ÷ Gamma. Theta and Gamma co-vary during working memory tasks. This ratio gives the forest indirect information about memory load.

These engineered features make the inter-band relationships **explicit** for the Random Forest, rather than leaving it to discover them implicitly.

Step 2: Temporal Smoothing

A **2-sample causal rolling mean** is applied. Each data point is replaced by the average of itself and the previous sample. 'Causal' means it only looks backward (no future data is used), making it fair. This reduces frame-to-frame jitter (random noise between consecutive seconds).

Step 3: MCAR Mask (Evaluation Only)

5% of values are set to NaN randomly (seed=42) to simulate realistic data loss. This is done **ONLY** for evaluation — to measure how well the imputer works on known missing values.

Step 4: StandardScaler Normalization

All columns are scaled to **zero mean and unit variance**. Without this, Delta band (mean=60, std=273) would dominate the forest's splitting decisions because its values are much larger than Gamma (mean=8, std=17). Scaling ensures all bands contribute equally.

Step 5: RF-MICE Imputation

scikit-learn's IterativeImputer runs with RandomForestRegressor as the base estimator. It runs up to 10 iterations, training a new random forest for each column with missing values in each iteration.

Step 6: Inverse Transform

The StandardScaler inverse transform converts the imputed values back to original $\mu\text{V}^2/\text{Hz}$ units.

Q34. What is StandardScaler? Why is it applied before and not inside the MICE loop?

StandardScaler is a preprocessing step that transforms each column to have **zero mean** and **unit variance** (standard deviation = 1). The formula is:

$$\text{scaled_value} = (\text{original_value} - \text{column_mean}) \div \text{column_std_deviation}$$

This ensures every column is on the same numerical scale. Without scaling, the Delta column (values up to 1000+ μV^2) would always be chosen as the splitting feature in decision trees because it has the largest numerical range — not because it is the most informative.

Applying **StandardScaler** **outside the MICE loop** (before all iterations, inverted after all iterations) ensures the same scale factors apply consistently across all 10 iterations. If it were inside the loop, the scale factors would shift with each iteration as missing values get filled in, causing **target distribution shift** — a known source of instability in iterative imputation.

Q35. What is a rolling mean? What does 'causal' mean in this context?

A **rolling mean** (also called a moving average) replaces each data point with the average of itself and nearby points. It smooths out short-term fluctuations while preserving longer-term trends.

A **2-sample rolling mean** means: $\text{new_value}[t] = (\text{value}[t] + \text{value}[t-1]) / 2$

Causal means the smoothing only uses **past and present** data — never future data. A non-causal (bidirectional) filter would look ahead in time, which is 'cheating' in time-series analysis: in real life, you cannot know future values when processing current data.

Why width = 2? A window of 1 does nothing. A window of 3+ introduces more smoothing but also more lag and uses more future data in edge cases. Width 2 is the minimal effective choice — it reduces most inter-frame jitter while remaining strictly causal.

SECTION 8: Results and Evaluation Metrics

Q36. What is RMSE? How do you interpret RMSE = 30.23?

RMSE (Root Mean Square Error) measures how far the imputed values are from the true values on average, with larger errors penalized more heavily.

Formula: $RMSE = \sqrt{[(1/n) \times \sum(\text{predicted} - \text{actual})^2]}$

RMSE = **30.23 $\mu V^2/Hz$** means that, on average, the imputed band-power values deviate from the true values by about 30.23 $\mu V^2/Hz$. The squaring step means large errors contribute much more than small ones — so RMSE is sensitive to outliers.

To judge whether 30.23 is good, consider the scale of the data: Delta band values can reach 1,000+ $\mu V^2/Hz$, so an error of 30 is relatively small. The R^2 score of 0.9861 confirms the error is small relative to the total variance.

Q37. What is MAE? How does MAE = 5.65 compare to RMSE = 30.23?

MAE (Mean Absolute Error) is the average of the absolute differences between predicted and actual values.

Formula: $MAE = (1/n) \times \sum|\text{predicted} - \text{actual}|$

MAE = **5.65 $\mu V^2/Hz$** . This is much smaller than RMSE (30.23), which tells us something important: **most errors are small (around 5–6 $\mu V^2/Hz$), but there are a few large outlier errors that inflate the RMSE.**

This pattern is typical in EEG data: most of the time imputation is very accurate, but occasionally a spike or burst in the signal is hard to predict, causing a large error for just that one point. MAE treats all errors equally; RMSE punishes large errors more.

Q38. What is R^2 Score? What does $R^2 = 0.9861$ mean?

R^2 (R-squared, or Coefficient of Determination) measures how much of the variance in the true values is explained by the imputed values. It ranges from 0 to 1 (sometimes negative for very bad models).

Formula: $R^2 = 1 - (SS_{\text{residual}} / SS_{\text{total}})$, where $SS_{\text{residual}} = \sum(\text{actual} - \text{predicted})^2$ and $SS_{\text{total}} = \sum(\text{actual} - \text{mean})^2$

- **$R^2 = 0$:** The model is no better than simply predicting the mean for every value.
- **$R^2 = 1$:** Perfect prediction — every imputed value exactly matches the true value.
- **$R^2 = 0.9861$:** The imputation explains **98.61% of the variance** in the missing values. Only 1.39% of the variance is unexplained. This is considered **excellent** for biomedical signal imputation.

Q39. What is Accuracy (98.61%) and how is it calculated here?

In the context of this regression imputation task, **Accuracy** = $(1 - \text{NRMSE}) \times 100\%$, where NRMSE is the Normalized RMSE (RMSE divided by the range or standard deviation of the data).

Accuracy = **98.61%** means the imputed values are, on average, within 1.39% of the true signal range from the actual values. This is a direct, intuitive interpretation of how close the imputed values are to the true values.

This high accuracy is achieved because:

- EEG band powers in the same session are highly correlated across time — knowing nearby values makes prediction easy.
- The engineered features (Alpha/Beta ratio, Theta/Gamma ratio) give the model explicit information about inter-band relationships.
- Random Forest is robust to outliers and captures non-linear patterns well.

Q40. Why was the evaluation done on 5% missing data? What if the missingness is higher?

5% was chosen because it realistically represents **typical electrode dropout rates** in a well-conducted EEG session. It is enough missing data to evaluate the imputer rigorously, but not so much that the imputation problem becomes extremely difficult.

The use of **strict MCAR conditions** (following Graham 2009 and Zhang 2015's framework) ensures the benchmark is not artificially inflated — some imputation papers use weaker evaluation protocols that make their results look better than they are.

For higher missingness rates (10%, 20%):

- At 10–15%, most well-designed imputation methods still work reasonably well if the remaining data is informative.
- Above 30%, imputation quality degrades significantly for any method because there is too little remaining signal to learn from.
- Future work could test HTI-MissForest at multiple missingness levels (5%, 10%, 20%) to characterize its robustness.

SECTION 9: Brain Diseases, Epilepsy, and Clinical EEG

Q41. What is epilepsy? How does EEG detect it?

Epilepsy is a chronic neurological disorder characterized by **recurrent, unprovoked seizures**. A seizure is a sudden burst of abnormal, excessive, or synchronous electrical activity in the brain.

Normally, neurons fire in controlled patterns. In epilepsy, groups of neurons suddenly fire all at once in an uncontrolled storm of activity. This disrupts normal brain function, causing seizures that can manifest as convulsions, loss of consciousness, unusual sensations, or staring spells.

How EEG detects epilepsy:

- **Inter-ictal spikes:** Between seizures, the epileptic brain shows characteristic sharp 'spike' discharges lasting 20–70 ms. These are the main diagnostic feature.
- **Spike-wave complexes:** Bursts of spikes followed by slow waves — classic for absence epilepsy.
- **Increased Delta activity:** During and after seizures, slow Delta waves dominate.
- **Abnormal synchrony:** Large, synchronized discharges from many electrodes simultaneously.

Q42. What are the different EEG spike/wave patterns seen in epilepsy?

Neurologists identify epilepsy type and severity from the **morphology (shape) of EEG patterns**. Common patterns include:

- **Spike:** A sharp pointed wave lasting less than 70 ms. Indicates focal irritability.
- **Sharp wave:** Similar to spike but lasting 70–200 ms. Less intense but same clinical significance.
- **Spike-and-wave complex:** A spike immediately followed by a slow wave. Classic pattern of generalized epilepsy.
- **Slow spike-and-wave:** Same pattern but at slower frequency (<2.5 Hz). Associated with Lennox-Gastaut syndrome (severe epilepsy).
- **Polyspike:** Multiple spikes in rapid succession. Associated with juvenile myoclonic epilepsy.
- **Polyspike-and-wave:** Multiple spikes followed by a slow wave. Seen in generalized tonic-clonic seizures.
- **Multiple sharp complexes:** Bursts of sharp waves from multiple brain regions simultaneously.

Q43. What is the difference between epileptic EEG and normal EEG?

The key visual differences between epileptic and normal EEG:

- **Amplitude:** Epileptic EEG shows high-amplitude signals (often >100 μ V, sometimes >200 μ V). Normal EEG is typically 20–100 μ V.
- **Regularity:** Epileptic EEG has periodic, stereotyped discharges that repeat rhythmically. Normal EEG is more irregular.
- **Spike morphology:** Sharp, pointed spikes are absent in normal EEG.
- **Synchrony:** During a generalized seizure, the ENTIRE scalp shows the same pattern simultaneously. In normal EEG, different regions show different patterns.
- **Frequency content:** Post-seizure EEG often shows heavy Delta (slow wave) activity — the brain is 'recovering' from the electrical storm.

Q44. What diseases are linked to which EEG patterns?

Different neurological and psychiatric conditions leave distinct fingerprints in EEG frequency patterns:

- **Epilepsy:** Spike discharges, increased Delta, spike-wave complexes
- **Alzheimer's Disease:** Diffuse slowing — increased Theta and Delta, decreased Alpha and Beta
- **ADHD:** Excess Theta, reduced Beta (especially the Theta/Beta ratio is used as a diagnostic aid)
- **Anxiety Disorder:** Elevated high Beta (>25 Hz), reduced Alpha
- **Depression:** Alpha asymmetry — lower Alpha in left frontal vs. right frontal regions
- **Alcohol Use Disorder:** Abnormal coherence between brain regions, disrupted Alpha and Beta patterns
- **Sleep Disorders:** Disrupted sleep stage transitions (normal Delta in deep sleep is replaced by lighter waves)
- **Schizophrenia:** Reduced Gamma, impaired Gamma synchrony during cognitive tasks

SECTION 10: Novelty, Future Scope, and General Concepts

Q45. What is the novelty of this project? How is it different from existing work?

This project makes **five distinct novel contributions** that, in combination, have not appeared in any prior published work:

- **First Arduino Mega IoT Grove Kit for EEG + imputation:** All existing low-cost EEG systems (OpenBCI, NeuroSky) use dedicated EEG hardware. This is the first to use a general-purpose IoT prototyping board and feed it directly into a validated imputation backend.
- **Physiologically motivated feature engineering:** Adding Alpha/Beta and Theta/Gamma ratio columns as input to MissForest is new. Previous MissForest applications used plain feature tables with no domain-specific engineering.
- **Separate treatment of the time column:** Using linear interpolation for the monotonically increasing time index (instead of putting it through Random Forest, which confuses tree splitters) is a deliberate and novel design choice.
- **Causal pre-smoothing:** The 2-sample causal rolling mean before MICE reduces imputation target variance without look-ahead — not seen in prior EEG imputation literature.
- **StandardScaler wrapped around (not inside) the MICE loop:** Prevents target distribution shift across iterations while ensuring Delta's large absolute values do not dominate the forest.

Q46. What is a Brain-Computer Interface (BCI)?

A **Brain-Computer Interface (BCI)** is a system that establishes a direct communication pathway between the brain and an external device, bypassing the body's normal neuromuscular pathways.

Imagine a patient with ALS (amyotrophic lateral sclerosis) who is completely paralyzed — they cannot move any muscle but their brain still works. A BCI reads their EEG signals, interprets their intended commands (e.g., 'move cursor left' vs. 'move cursor right' based on motor imagery brain patterns), and sends those commands to a computer or robotic device.

BCI applications:

- Communication aids for locked-in syndrome patients
- Prosthetic limb control for amputees
- Wheelchair control for quadriplegic patients
- Gaming and virtual reality
- Neurofeedback training for ADHD and anxiety

For BCI to work reliably, the EEG signal must be **clean and gap-free**. This is exactly what HTI-MissForest provides — removing missing data before the BCI classifier trains on it.

Q47. What is neurofeedback? How does it use EEG?

Neurofeedback is a type of biofeedback therapy where patients learn to regulate their own brain activity by watching real-time EEG data. It is based on **operant conditioning** — rewarding the brain when it produces desired patterns.

For example, in ADHD neurofeedback:

- The patient plays a simple video game controlled entirely by their brain waves.
- When their Beta power increases (focused attention) and Theta power decreases, the game rewards them (the character speeds up, a star appears).
- Over many sessions (typically 30–40), the brain learns to maintain the desired state automatically.

Clinical research shows neurofeedback can reduce ADHD symptoms comparably to medication in some patients, without side effects. It is also used for anxiety, PTSD, and peak performance training in athletes.

Q48. What is MissForest? How does this project improve upon it?

MissForest was introduced by Stekhoven and Buhlmann in 2012 as a general-purpose imputation method for mixed-type data (both numerical and categorical columns). It uses Random Forest in an iterative loop — essentially the same MICE framework but with Random Forest as the regression engine instead of linear regression.

It was shown to outperform KNN, mean imputation, and multiple imputation methods on various medical datasets. However, MissForest was designed for **general-purpose tabular data**, not specifically for EEG.

How HTI-MissForest improves upon original MissForest:

- **EEG-specific feature engineering:** Original MissForest uses raw feature columns. HTI-MissForest adds physiologically motivated ratio features derived from EEG neuroscience.
- **Temporal smoothing:** Original MissForest treats each row independently. HTI-MissForest acknowledges that EEG is a time series and applies causal smoothing to exploit temporal correlations.
- **Normalization strategy:** The StandardScaler-outside-loop approach prevents distribution shift, a problem not addressed in the original.

Q49. What is Google Colab and why was it used?

Google Colab (Colaboratory) is a free, cloud-based Python programming environment provided by Google. It runs in a web browser and provides free access to:

- Python 3 runtime with most scientific libraries pre-installed
- Free GPU (Graphics Processing Unit) for faster machine learning
- Free T4 GPU (NVIDIA Tesla T4) — enough for this project's computational needs
- Easy sharing via Google Drive

Using Colab makes this project **fully reproducible by anyone, anywhere, for free**. A researcher in a rural clinic with only a laptop and internet connection can run the entire pipeline without installing anything or paying for software licenses.

This aligns with the project's core goal: making EEG data collection and processing accessible in resource-limited settings.

Q50. What are the future scope and next steps after data collection and imputation?

This project covers the first two critical steps: **data collection** and **data imputation**. The full pipeline for a clinical EEG application requires several more steps:

Immediate Next Steps:

- **Brain-state classification:** Use the clean, imputed dataset to train classifiers (SVM, Random Forest, CNN) to distinguish between the 4 brain states (Learning, Maths, Meditation, Song Listening).
- **Disorder detection:** Apply classification to the BRMH Kaggle dataset to classify patients by disorder (epilepsy, ADHD, depression, etc.).

Longer-Term Future Scope:

- **Multi-channel EEG:** Extend to 16–32 channels for richer spatial brain mapping and improved classification accuracy.
- **Deep Learning Imputation:** Use LSTM or Transformer models that are inherently time-series-aware, potentially outperforming tree-based MICE for long recordings.
- **Wireless transmission:** Activate the onboard SIM7672S cellular modem to transmit recordings directly to a cloud server for telemedicine applications.
- **Real-time classification:** Integrate the entire pipeline (collection → filter → impute → classify) into a live streaming system.
- **Clinical validation:** Test on TUH epilepsy dataset and collaborate with hospitals for prospective patient validation.
- **Edge deployment:** Compress and run the imputation model on a Raspberry Pi 4 for fully offline, on-device inference in areas with no internet.